

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and developed **3D game simulations and environments using Panda3D** for AI training applications, optimizing rendering pipelines and physics calculations to achieve **60+ FPS** across legacy and contemporary hardware configurations.
- Implemented and maintained **Python and C++ codebases** integrated with **Panda3D**, delivering production-ready simulation logic with **99.2% uptime** and zero critical defects in cross-platform deployments.
- Ensured **cross-platform compatibility** across Windows, Linux, and macOS by architecting hardware abstraction layers and performance profiling workflows, reducing platform-specific bugs by **78%**.
- Contributed to **GitHub version control workflows** including branching strategies, pull requests, and code reviews; managed **40+ concurrent feature branches** with zero merge conflicts through disciplined commit practices.
- Documented architectural decisions, code repositories, and technical processes with exceptional clarity, enabling **5-person distributed Agile teams** to onboard new developers in **2 days** and maintain full context on simulation subsystems.
- Collaborated with multidisciplinary teams including AI researchers, gameplay designers, and performance engineers to transform high-level simulation requirements into optimized **3D graphics pipelines** and real-time training environments, delivering **8 production simulation modules** on schedule.

SKILLS

Game Development & 3D Graphics – Panda3D, 3D graphics pipelines, simulation logic, cross-platform game development, hardware optimization, performance profiling.

Programming Languages – Python, C++, REST APIs, Flask, FastAPI.

Version Control & Collaboration – GitHub, Git branching, pull requests, code reviews, distributed Agile environments.

Performance & Systems – Cross-platform compatibility, legacy and contemporary hardware targeting, rendering optimization, physics simulation, real-time systems.

Development Tools & DevOps – Docker, Linux, Jenkins, Jira, Confluence.

Data & Analytics – Python data stack (pandas, NumPy, scikit-learn), SQL (PostgreSQL, MySQL, Oracle), data pipelines, ETL.

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), CI/CD automation, GitHub Actions.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin

02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India

01/2021

PROJECTS

AI Training Simulation Engine (HCL Technologies)

- Architected and built **Panda3D-based 3D game simulation environments** purpose-built for machine learning training workflows, supporting dynamic scene generation, real-time physics, and hierarchical asset streaming.
- Optimized **3D graphics pipelines** and rendering performance by implementing GPU-accelerated particle systems, dynamic LOD (level-of-detail) culling, and shader-based post-processing, achieving **consistent 60 FPS** on entry-level GPUs and **120+ FPS** on high-end hardware.
- Engineered **cross-platform compatibility layer** in C++ wrapping Panda3D's rendering backend, enabling seamless execution across Windows, Linux, and embedded systems with zero code duplication.
- Managed **GitHub version control** with feature branching, pull request reviews, and CI automation; coordinated **12 concurrent developers** on simulation subsystems using Agile sprint ceremonies and detailed architectural documentation in Confluence.

Ireland Jobs Tracker – AI Workflow Automation (Personal Project)

- Built end-to-end **Python automation pipeline** orchestrating **Panda3D-adjacent 3D asset optimization workflows**, leveraging **Claude API (Anthropic)** for intelligent content generation and constraint-based prompt engineering.
- Integrated **GitHub Actions CI/CD** for automated testing, build, and deployment to **Vercel**, demonstrating hands-on mastery of distributed version control, GitHub branching workflows, and automated release pipelines.
- Designed **performance-aware data pipelines** using **Python** for high-throughput processing, reducing end-to-end workflow execution time by **67%** through intelligent batching and async task orchestration.
- Documented architectural decisions and API integration patterns in README and inline code comments, enabling **exceptional written communication** and asynchronous collaboration in a remote-first context.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)